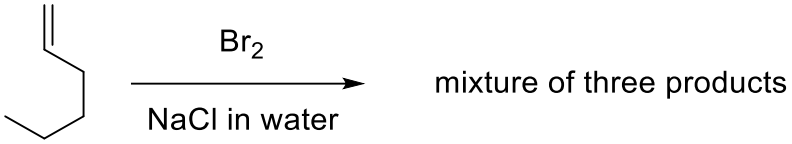
1. What are the chemical structures of the three possible products? Explain the regioselectivity.

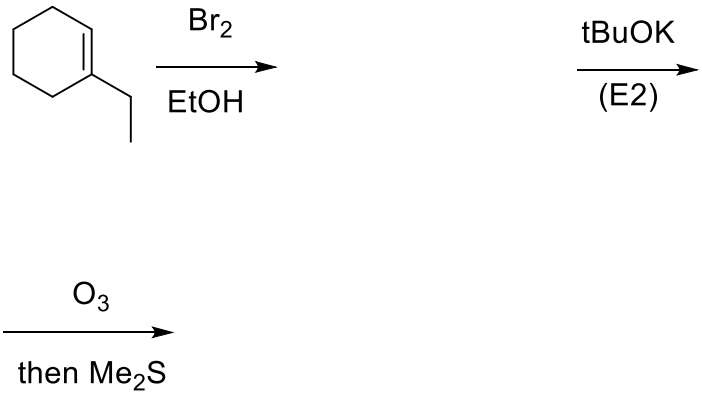


2. What are the chemical structures of the products for the following 4-step transformation? Explain the stereochemistry of the products

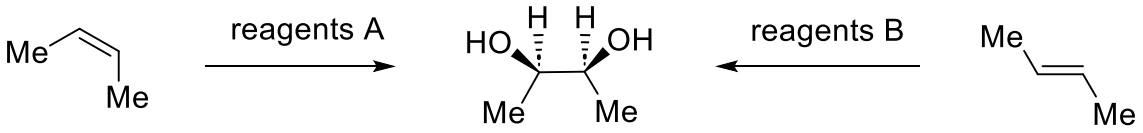




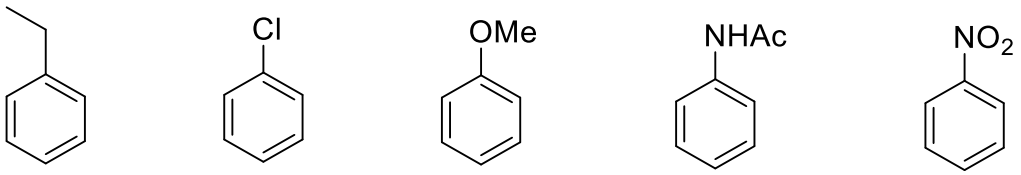
3. What are the chemical structures of the products for the following 3-step transformation?



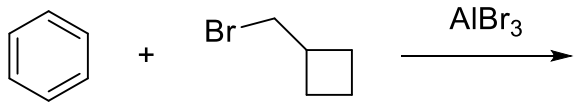
4. What are the reaction reagents for the following transformations? (The transformations might involve more than one step.)



5. Predict the major products of reactions when each of the following reacts with (1) bromine and $FeBr_3$ (2) HNO_3 and H_2SO_4 (conc.).



6. Predict the reaction product. Provide a mechanism for the following reaction.



product containing a 5-membered ring

7. Propose a synthetic plan for the following compound.

